Banaan Kiamanesh

Robotics and Control Systems Engineer

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Profile

Control Systems Engineer and Roboticist completing an M.Sc. at the University of Padova. Turns control theory into reliable, field-ready systems across aerial robotics, embedded control, sensor fusion, manipulators and networked systems. Develops end-to-end experiments from modelling and simulation to bench and vehicle tests, with reproducible software, measured performance and clear documentation.

Technical Stack

Programming	MATLAB/Simulink, Python, C/C++, Julia, PyTorch, NumPy, Git, Linux
Robotics	ROS 2, Simscape Multibody, PyBullet, URDF, Dynamics/Control, Path Planning, Multi-Agent Systems, Deep-Learning and Reinforcement Learning
Control Systems	Linear Control, Digital Control, Non-Linear Control, Robust Sliding-Mode, MPC, RL
Estimation	System Identification, EKF/ESKF, AHRS, Sensor Fusion
Embedded Systems	STM32, Raspberry Pi, Arduino, AVR, Sensors and Actuators

Work Experience

Autonomous Driving Team Member **2026**
UniPd DriveOps *Padova, Italy*

- Contribute to the University of Padova team competing in the Bosch Future Mobility Challenge; placed 13th in the 2026 semifinal phase.

Digital Signal Processing R&D Engineer **08/2023 – 02/2024**
iEEG.co *Remote*

- Developed feature-engineering and classification pipelines for SSVEP and motor-imagery EEG; supported an EEG-controlled wheelchair project.

R&D Intern, EEG Signal Processing **03/2023 – 06/2023**
iEEG.co *Remote*

- Designed filters for raw EEG signals and evaluated denoising in the time and frequency domains.

Projects

Icarus – Quadcopter Dynamics and Control **2023**
Bachelor's thesis

- Derived a 6-DoF model and implemented PID, optimisation-based multi-flips and robust sliding-mode control.

QuadCopter NLPNN Constrained MPC **2026**
Python, adaptive neural networks and MPC

- Implemented a constrained MPC using an adaptive nonlinear-in-parameter neural network observer/identifier for prediction, with a 20-step horizon and state/input constraints.

ManiDyn – Symbolic Robot Dynamics Toolbox **2025 – 2026**
Open-source MATLAB toolbox

- A MATLAB Toolbox for Symbolic Calculation of Serial Link Manipulator Kinematics and Dynamics, directly from the URDF File and Generating Accelerated MEX/C File.

Kinova Gen3 Model-Based Control **2025**
MATLAB, Simulink and Simscape

- Compared six 6-DoF task-space controllers, including feedback linearisation, inverse dynamics, sliding-mode and robust control.

Dual-IMU Sensor Fusion and Attitude Estimation **2026**
MATLAB and Python

- Comparison of EKF, ESKF and Mahony Filters on a Dual IMU Sensor Fusion Task.

Para-Seq – Parallel Sequence Computing **2026**
PyTorch and CUDA

- Parallelization of RNN Models and ODE Solvers.

WayPoint – Autonomous Drone Mission Planner **2026**
Browser-based robotics tool

- A Website for Mission Planning for DJI Drones, Producing KMZ/KML/CSV Files Based on a Given Map Area.

Self-Triggered and Event-Triggered Control **2025**
Networked control

- Implemented state-dependent update strategies for reducing computation and communication under disturbance and noise.

SIS Parameter Estimation **2025**
Nonlinear estimation

- Applied EKF and MCMC to estimate parameters of a nonlinear susceptible–infected–susceptible model.

Minimum-Grad – Automatic Differentiation Engine 2025
Python and NumPy

- Built tensor operations, computation graphs, reverse-mode gradients, optimisers and small neural-network examples.

Optimization4ML 2025
Numerical optimisation

- Implemented gradient descent, RMSProp and Adam to study optimiser dynamics and convergence.

JigSaw OpenCV 2025
Classical computer vision

- Solved image-puzzle reconstruction tasks with classical OpenCV processing and geometric matching.

Flocking and Multi-Agent Control 2024 – 2025
MATLAB and PyBullet

- Implemented Boids and Olfati-Saber algorithms with tracking, obstacle avoidance and quadrotor-swarm extensions.

Fault-Tolerant Cooperative Aerial Load Transport 2024
Academic project *Cooperative control*

- Combined Euler–Lagrange modelling, sliding-mode observation, adaptive control, PSO trajectories and rope-failure handling.

Neural Observer and Identifier for Nonlinear MIMO Systems 2024
Recurrent neural networks

- Implemented a learned state observer and system identifier for nonlinear Lorenz-system dynamics.

Sliding-Mode Control Bible 2024
Nonlinear control

- Implemented reusable examples from nonlinear-control literature for plants, observers and robotic systems.

Path Planning Algorithms 2024
Autonomous systems

- Implemented ACO, GA, PSO and Q-learning planners with spline-based path smoothing.

Mass-Spring-Damper Identification and Control 2023
Industrial control project

- Performed grey-box identification and compared manual, toolbox and optimisation-based PID tuning.

Deep Learning for Navigation 2022
IMU navigation estimation

- Logged 9-DoF IMU data at 100 Hz and compared LSTM, KNN and SVR navigation estimators.

Evo-Copter 2022
Raspberry Pi flight control *Portfolio project*

- Built a Raspberry Pi Quadcopter Controller and refined PID gains using biogeography-based optimisation.

Pose-Ordered Tello – Vision-Based Drone Control 2022
Computer vision and aerial robotics

- Combined MediaPipe pose estimation, person tracking and decision-tree classification with DJI Tello commands.

Raspi-Kapta – Embedded Flight Controller 2019 – 2022
Raspberry Pi and C++

- Integrated radio input, BNO055 sensing and motor PWM with anti-windup PID, calibration and safety tools.

Low-Cost Radio and Embedded Communications 2020 – 2021
AVR, NRF24L01 and Processing *Embedded systems*

- Developed transmitter/receiver firmware, a Processing ground-station interface and a manufacturable PCB.

Education

M.Sc. Control Systems Engineering – Robotics 2024 – Present
University of Padova *Padova, Italy*

- Thesis: "Parallelization of Sequential Models: from RNN Models to Theorization of Parallel Carleman Based NMPC"

B.Sc. Electrical Engineering – Control Systems and Dynamics 2023
University of Tabriz *Tabriz, Iran*

- Thesis: "Design and Implementation of a Robust Non-Linear Sliding-Mode Controller for Quadcopter + High-Speed Quadcopter Flips"

Languages

English	C1 – TOEFL iBT 102 (R26, L29, S23, W24)
Persian	C2
Azeri	Mother Tongue
Turkish	C1

Certifications

- **Robotics Specialization** – University of Pennsylvania
- **Non-Invasive Brain Stimulation** – National Brain Mapping Lab
- **BCI & Neurotechnologies Spring School** – g.tec Medical Engineering
- **Mathematical Psychology: Deep Learning and Neural Networks** – Shahid Beheshti University
- **Mathematics for Machine Learning** – Imperial College London
- **MATLAB Programming for Engineers and Scientists** – Vanderbilt University
- **Signal Processing: EEG Signals** – onlinebme.com
- **A1/A3 Drone Operating License - EASA**